

ULRICHEN, Switzerland

WMO: 067450

Lat: 46.5049N

Lon: 8.3082E

Elev: 4420

StdP: 12.50

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -1.5	(d) -10.5	(e) 3.7	(f) -2.6	(g) -7.0	(h) 4.4	(i) 0.7	(j) 20.5	(k) 27.1	(l) 17.5	(m) 26.7	(n) 1.0	(o) 70	(p) 0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 24.1	(c) 79.5	(d) 59.4	(e) 76.5	(f) 58.5	(g) 73.7	(h) 57.3	(i) 61.1	(j) 75.7	(k) 59.7	(l) 73.2	(m) 58.4	(n) 70.5	(o) 8.5	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.5	80.2	62.3	55.1	76.1	61.6	53.7	72.4	61.1	29.3	75.8	28.3	73.4	27.3	70.8	66.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.7	(b) 17.0	(c) 14.3	(e) -12.2	(f) 84.2	(g) 6.3	(h) 2.8	(i) -16.7	(j) 86.2	(k) -20.3	(l) 87.9	(m) -23.9	(n) 89.4	(o) -28.4	(p) 91.5		
(4) 20.7	(5) 17.0	(6) 14.3	(7) -12.5	(8) 63.4	(9) 6.2	(10) 1.4	(11) -16.9	(12) 64.4	(13) -20.5	(14) 65.3	(15) -24.0	(16) 66.1	(17) -28.5	(18) 67.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.5	19.5	22.9	31.6	40.4	47.5	55.7	58.8	57.7	51.4	44.4	32.4	22.2
	DBStd	15.29	8.40	8.09	6.90	5.55	5.65	5.59	4.96	5.15	5.27	6.27	7.80	8.61
	HDD50	4327	944	760	572	290	115	14	3	7	47	186	528	861
	HDD65	8963	1409	1180	1037	738	543	281	196	229	409	638	977	1326
	CDD50	849	0	0	0	2	37	185	277	247	88	13	0	0
	CDD65	9	0	0	0	0	0	2	4	3	0	0	0	0
	CDH74	573	0	0	0	0	4	149	221	185	14	0	0	0
	CDH80	69	0	0	0	0	0	19	21	29	0	0	0	0
Wind	WSAvg	3.9	2.4	3.1	4.1	5.2	5.4	5.1	4.8	4.3	4.1	3.3	2.6	2.1
Precipitation	PrecAvg	62.2	3.9	4.1	4.1	4.9	6.3	6.1	5.5	6.1	5.4	5.0	6.3	4.5
	PrecMax	80.4	9.5	11.6	8.0	9.1	14.1	10.2	10.0	10.8	13.3	11.9	19.4	11.5
	PrecMin	43.5	0.9	0.8	0.7	0.8	1.8	2.4	3.0	1.9	1.6	0.6	1.2	0.5
	PrecStd	8.2	2.3	2.5	2.1	2.2	3.1	1.9	1.8	1.9	2.8	2.7	4.0	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	45.2	54.8	65.6	73.8	82.3	82.5	83.6	75.9	67.9	56.4	42.2
		MCWB	34.8	35.7	40.4	47.8	55.0	59.7	61.1	59.9	56.3	52.5	44.1	35.8
	2%	DB	36.1	41.2	51.0	61.4	68.8	78.3	79.1	78.8	71.8	63.8	51.0	37.9
		MCWB	33.4	33.9	39.4	46.2	52.4	59.0	59.3	59.7	54.9	50.0	42.1	34.0
	5%	DB	33.9	37.6	47.5	57.3	64.8	74.7	76.3	75.2	68.3	60.3	46.9	35.2
		MCWB	32.3	32.2	37.4	44.0	50.6	57.7	58.5	58.4	53.7	48.7	40.4	33.2
	10%	DB	32.3	34.5	43.7	53.4	60.9	71.0	73.3	71.8	64.7	56.6	42.8	33.5
		MCWB	31.0	31.4	35.7	42.2	48.5	56.1	57.4	57.3	52.2	46.9	39.3	32.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	37.4	42.1	49.2	55.8	62.1	62.8	62.6	58.5	54.2	46.9	37.6
		MCDB	37.6	42.4	51.9	63.3	70.7	78.0	78.2	77.6	70.2	64.5	51.9	40.6
	2%	WB	34.1	34.9	40.1	46.8	53.6	60.2	61.1	60.8	56.6	52.0	43.7	35.1
		MCDB	35.2	39.1	48.8	59.7	65.6	75.3	75.8	74.6	67.6	59.8	48.8	36.6
	5%	WB	32.6	33.3	38.3	44.9	51.6	58.5	59.7	59.4	55.0	50.2	41.5	33.7
		MCDB	33.7	36.5	45.9	55.8	62.2	72.3	73.1	71.8	65.4	57.0	46.0	34.9
	10%	WB	31.2	31.9	36.4	42.7	49.7	56.8	58.3	58.0	53.4	48.5	38.9	32.4
		MCDB	32.5	34.3	42.3	51.9	59.0	68.8	70.2	69.1	62.2	54.7	42.7	33.6
Mean Daily Temperature Range	5% DB	MDBR	19.0	21.9	21.0	20.4	20.6	23.9	24.1	23.3	23.0	21.7	16.6	16.7
		MCDBR	16.4	25.3	27.8	29.2	29.3	32.4	31.3	32.0	31.1	28.6	23.1	16.5
		MCWBR	13.7	18.6	16.6	15.0	14.4	14.8	13.3	14.3	15.2	16.0	14.4	12.6
	5% WB	MCDBR	14.2	20.9	23.4	26.8	25.9	29.8	28.3	27.7	26.6	23.7	20.3	13.1
		MCWBR	12.1	15.6	14.0	14.2	13.1	13.8	12.5	12.9	13.3	13.7	13.2	10.5
Clear-Sky Solar Irradiance	taub	0.252	0.281	0.331	0.382	0.403	0.410	0.407	0.395	0.366	0.334	0.288	0.250	
	taud	2.290	2.213	2.129	2.099	2.159	2.197	2.207	2.257	2.322	2.381	2.389	2.326	
	Ebn at Noon	270	282	281	276	274	272	271	269	267	260	254	258	
	Edh at Noon	24	32	41	47	46	45	43	40	34	28	22	21	
All-Sky Solar Radiation	RadAvg	497	801	1180	1411	1525	1799	1793	1557	1217	805	491	413	
	RadStd	61	96	170	224	149	162	156	139	117	93	72	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	+5.60	+6.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (119 neighbors)	N/A	N/A	N/A	+1.31	+0.76	+0.58	N/A	N/A	+104	+29

Nomenclature: See separate page